A2 · Unit 5 · WCH15 Transition Metals and Organic Nitrogen Chemistry

Organic nitrogen compounds

Nitrogen's lone pair is the key to this topic. It makes amines basic, lets them act as ligands and nucleophiles, and joins monomers into nylon, Kevlar and proteins. You will also meet azo dyes, the zwitterions of amino acids, and the difference between addition and condensation polymers.

Naming amines and amides

- **Primary amines** have the –NH_2 group. $CH_3CH_2CH_2NH_2$ is called propylamine (the name the exam board often uses) or propan-1-amine (the IUPAC name, built like an alcohol name). With a higher-priority group present, the prefix **amino-** is used: $CH_3CH(NH_2)COOH$ is 2-aminopropanoic acid.
- **Secondary and tertiary amines**: the longest chain gives the name and other alkyl groups on nitrogen are given the locant N: $CH_3CH_2CH_2NHCH_3$ is N-methylpropylamine (N-methylpropan-1-amine); $(CH_3)_2NCH_2CH_2CH_3$ is N,N-dimethylpropylamine. If both groups are the same, $(CH_3CH_2)_2NH$ is diethylamine.
- **Diamines**: $H_2N(CH_2)_6NH_2$ is hexane-1,6-diamine.
- **Amides** end in -amide: CH_3CONH_2 is ethanamide; $CH_3CH_2CONHCH_3$ is N-methylpropanamide; $CH_3CH_2CON(CH_3)_2$ is N,N-dimethylpropanamide.

Amines as bases

Small amines smell fishy and dissolve readily in water, because the N–H groups and the nitrogen lone pair hydrogen bond with water molecules. The lone pair on nitrogen can accept a proton, so amines are **Brønsted–Lowry bases**:

$$CH_3NH_2(aq) + H_2O(l) \rightleftharpoons CH_3NH_3^+(aq) + OH^-(aq)$$

How strong a base an amine is depends on how available its lone pair is:

- **Aliphatic amines are stronger bases than ammonia**: alkyl groups are electron-releasing, pushing electron density onto the nitrogen and making its lone pair more available. Secondary amines, with two alkyl groups, are stronger still: diethylamine > ethylamine > ammonia.
- **Phenylamine is a much weaker base than ammonia**: its lone pair is partly delocalised into the benzene ring, so it is less available to accept a proton. Its solutions are barely alkaline.
- **Amides are not basic**: the nitrogen lone pair is delocalised towards the electronegative C=O group.

Salts, buffers and complexes

- **With acids** amines form ammonium salts: $CH_3NH_2 + HCl \rightarrow CH_3NH_3^+Cl^-$ (methylammonium chloride); $2CH_3NH_2 + H_2SO_4 \rightarrow (CH_3NH_3^+)_2SO_4^{2-}$. These ionic salts are crystalline solids that dissolve in water even

when the amine does not: phenylamine is only slightly soluble, but phenylammonium chloride is soluble. Adding NaOH turns the salt back into the amine.

- **Basic buffers** can be made from an amine and its salt, such as methylamine with methylammonium chloride, just as with ammonia and ammonium chloride.
- **As ligands**, amines form complexes with transition metal ions, like ammonia does: $[Cu(H_2O)_6]^{2+} + 4CH_3CH_2NH_2 \rightarrow [Cu(CH_3CH_2NH_2)_4(H_2O)_2]^{2+} + 4H_2O$, a deep blue solution.

Amines as nucleophiles

With halogenoalkanes

The lone pair also makes amines nucleophiles. A primary amine attacks a halogenoalkane in nucleophilic substitution to give a secondary amine; a second amine molecule removes a proton from the intermediate:

$$2CH_3CH_2NH_2 + CH_3CH_2Br \rightarrow (CH_3CH_2)_2NH + CH_3CH_2NH_3^+Br^-$$

The product still has a lone pair, so it can react again, giving a tertiary amine and finally a **quaternary ammonium salt**, $R_4N^+X^-$, in which the nitrogen has no lone pair left. Mixtures of products are normal.

With acyl chlorides

Amines react with acyl chlorides at room temperature to give N-substituted amides (an addition–elimination reaction, see Topic 15): $CH_3COCl + 2CH_3NH_2 \rightarrow CH_3CONHCH_3 + CH_3NH_3^+Cl^-$. Phenylamines react in the same way; paracetamol is made by acylating the $-NH_2$ group of 4-aminophenol with an ethanoyl group.

Making amines

Method	Reagents and conditions	Comment
halogenoalkane + ammonia	excess ammonia dissolved in ethanol, heated in a sealed tube: $CH_3CH_2Br + 2NH_3 \rightarrow CH_3CH_2NH_2 + NH_4Br$	one step, but the amine formed reacts further, giving a mixture of amines. A large excess of ammonia makes the primary amine the main product.
via a nitrile	1. KCN in aqueous ethanol, heat under reflux: $CH_3CH_2Br + CN^- \rightarrow CH_3CH_2CN + Br^-$. 2. $LiAlH_4$ in dry ether (or H_2 with a nickel catalyst): $CH_3CH_2CN + 4[H] \rightarrow CH_3CH_2CH_2NH_2$	gives only the primary amine, with one extra carbon; but it is two steps (lower overall yield) and KCN is highly toxic

reducing a nitroarene	tin (or iron) and concentrated hydrochloric acid, heat: $C_6H_5NO_2 + 6[H] \rightarrow C_6H_5NH_2 + 2H_2O$	in the acid the product is the salt $C_6H_5NH_3^+Cl^-$; add NaOH to release phenylamine, then separate it by steam distillation

Azo dyes

Making an azo dye from phenylamine

1. **Diazotisation**: react phenylamine with nitrous acid below 10 °C. Nitrous acid is unstable, so it is made in the flask from sodium nitrite and hydrochloric acid: $NaNO_2 + HCl \rightarrow HNO_2 + NaCl$.

2. $C_6H_5NH_2 + HNO_2 + HCl \rightarrow C_6H_5N_2^+Cl^- + 2H_2O$, forming benzenediazonium chloride, which contains the $–N{\equiv}N^+$ group.

   Keep it cold: above about 10 °C the diazonium ion reacts with water, $C_6H_5N_2^+ + H_2O \rightarrow C_6H_5OH + N_2 + H^+$, releasing nitrogen gas. Much below 0 °C the reaction is too slow.

3. **Coupling**: add the diazonium salt to phenol dissolved in sodium hydroxide. The diazonium ion is a weak electrophile that attacks the activated ring: $C_6H_5N_2^+ + C_6H_5OH \rightarrow C_6H_5–N{=}N–C_6H_4OH + H^+$.

4. The product, 4-hydroxyazobenzene, is a yellow-orange **azo dye**. The –N=N– link joins two benzene rings into one long delocalised system that absorbs visible light. Azo compounds are used as dyes, pigments and food colourings.

Polymers: addition and condensation

Addition polymers

Monomers with a C=C double bond join without losing any atoms. Draw the monomer with its other groups arranged around the double bond, then open the double bond to form the repeat unit. Propenamide, $CH_2{=}CHCONH_2$, gives poly(propenamide), $–[CH_2–CH(CONH_2)]–$. Poly(ethenol), $–[CH_2–CH(OH)]–$, has an –OH group on every other carbon, so it forms many hydrogen bonds with water and dissolves: it is used for soluble laundry bags and detergent capsules. Most poly(alkenes), with only strong non-polar C–C and C–H bonds, are unreactive and not biodegradable.

Condensation polymers: polyamides

In condensation polymerisation, two different functional groups react and a small molecule (water or HCl) is lost at each link. A **polyamide** forms from a dicarboxylic acid (or diacyl dichloride) and a diamine, linked by **amide groups**, –CONH–:

- **Nylon 6,6**: hexanedioic acid + hexane-1,6-diamine. Repeat unit $–CO(CH_2)_4CONH(CH_2)_6NH–$. The "6,6" counts the six carbons in each monomer.
- **Kevlar**: benzene-1,4-dicarboxylic acid + benzene-1,4-diamine. Repeat unit $–CO–C_6H_4–CONH–C_6H_4–NH–$. Its flat chains line up and hydrogen bond strongly, making it extremely strong: it is used in body armour.
- One monomer with an amine group at one end and an acid group at the other can also polymerise on its own, as amino acids do to make proteins.
- Asked for the *type of polymer*, say polyamide or polyester; asked for the *type of polymerisation*, say condensation.

- **Intermolecular forces**: polyamide chains hydrogen bond to each other between the N–H of one chain and the C=O of another, as well as having dipole–dipole and London forces. Polyesters have no N–H, so only dipole–dipole and London forces act between chains; polyamides therefore have higher melting points.
- **Hydrolysis**: the polar amide (and ester) links can be attacked, so condensation polymers are biodegradable. Acid hydrolysis of nylon 6,6 gives hexanedioic acid and the diammonium salt of the diamine ($H_3N^+(CH_2)_6NH_3^+$); alkaline hydrolysis gives the diamine and the salt of the acid ($^-OOC(CH_2)_4COO^-$).

Amino acids

An α-amino acid has an amine group and a carboxylic acid group on the same carbon, $H_2NCH(R)COOH$. Glycine, the simplest, has R = H: aminoethanoic acid. Others are named systematically, for example 2-amino-3-hydroxypropanoic acid (R = CH_2OH) or 2-aminobutanedioic acid (aspartic acid, with a second –COOH in the R group, an acidic amino acid). Lysine has a second $–NH_2$ and is a basic amino acid.

- **Optical activity**: every amino acid except glycine has four different groups on the central carbon, so it is chiral and its enantiomers rotate plane-polarised light in opposite directions. Glycine has two hydrogens on that carbon.
- **Zwitterions**: in the solid and in neutral solution, the acid group donates its proton to the amine group, giving a **zwitterion**, $^+H_3NCH(R)COO^-$, with a positive and a negative charge but no overall charge. Strong ionic attractions between zwitterions give amino acids their high melting points, far above those of molecules held only by hydrogen bonds.
- **Acid–base behaviour**: in acid the COO^- group gains a proton, giving the cation $^+H_3NCH(R)COOH$; in alkali the NH_3^+ group loses one, giving the anion $H_2NCH(R)COO^-$. With glycine: $^+H_3NCH_2COO^- + HCl \rightarrow {}^+H_3NCH_2COOH\ Cl^-$, and $^+H_3NCH_2COO^- + NaOH \rightarrow H_2NCH_2COO^-Na^+ + H_2O$. Because they can take up both, amino acid solutions act as weak buffers. Extra acid or amine groups in the R group change state too.
- Their functional groups also react normally: for example, heating with methanol and an acid catalyst esterifies the –COOH group.

Peptides and proteins

- The –COOH of one amino acid and the $–NH_2$ of another condense, losing water, to form a **peptide link**, –CONH– (an amide group). Two amino acids give a **dipeptide**; because either one can provide the –COOH, two different amino acids can form **two** different dipeptides.
- **Proteins** are long polypeptide chains, condensation polymers of amino acids.
- The chains fold into spirals (α-helices) and sheets held in shape by **hydrogen bonds** between the N–H of one peptide link and the C=O of another. This three-dimensional shape is vital to a protein's job, such as an enzyme's active site.
- Heating a protein with 6 mol dm^{-3} hydrochloric acid under reflux hydrolyses the peptide links, releasing the amino acids, which can then be identified by chromatography with ninhydrin (Topic 15).

Quick checks

QUICK CHECK

Which is the strongest base?

- ammonia
- phenylamine
- diethylamine
- ethanamide

Two electron-releasing ethyl groups make diethylamine's lone pair the most available. Phenylamine's lone pair is delocalised into the ring and ethanamide's towards the C=O, so both are much weaker.

QUICK CHECK

Why is a large excess of ammonia used to make ethylamine from bromoethane?

- ammonia is a catalyst
- so that bromoethane is more likely to meet ammonia than the ethylamine already formed, reducing further substitution
- to neutralise the HBr formed
- ammonia is the solvent

Ethylamine is itself a nucleophile and can react with more bromoethane to give secondary and tertiary amines. Swamping it with ammonia makes the primary amine the main product.

QUICK CHECK

Why must the diazotisation of phenylamine be kept below 10 °C?

- nitrous acid boils at 10 °C
- the azo dye decomposes
- above 10 °C the diazonium ion reacts with water to form phenol and nitrogen gas
- phenylamine freezes at higher temperatures

Benzenediazonium ions are unstable when warm. The low temperature keeps them long enough to couple with phenol.

QUICK CHECK

How many molecules of water are released when 50 molecules of hexanedioic acid react with 50 molecules of hexane-1,6-diamine to form one nylon 6,6 chain?

- 50
- 99
- 100
- 101

100 monomers joined in a single chain need 99 amide links, each releasing one water molecule. (For large n, 2n − 1 is usually written as 2n.)

QUICK CHECK

Why do amino acids have high melting points?

- they are large molecules with strong London forces
- they form covalent networks
- they exist as zwitterions, held together by strong ionic attractions
- they hydrogen bond more than water

In the solid, each molecule carries NH_3^+ and COO^- groups, so the crystal is held together like an ionic lattice.

QUICK CHECK

Which form of glycine is present at pH 1?

- $H_2NCH_2COO^-$
- H_2NCH_2COOH
- $^+H_3NCH_2COOH$
- $^+H_3NCH_2COO^-$

In strong acid both groups are protonated, giving the cation. H_2NCH_2COOH, with no charges, is never the main form in solution.

QUICK CHECK

Why do polyamides usually have higher melting points than polyesters of similar chain length?

- polyamides have stronger covalent bonds in the chain
- polyamide chains hydrogen bond to each other through N–H and C=O groups
- polyesters have no permanent dipoles
- polyamides are addition polymers

Both have C=O dipoles, but only polyamides have N–H groups able to hydrogen bond to C=O groups in neighbouring chains.

QUICK CHECK

Glycine (G) and alanine (A) are joined by a single peptide link. How many different dipeptides can form from one of each?

1

2

3

4

Gly–Ala and Ala–Gly are different: either amino acid can provide the –COOH that forms the peptide link. (Allowing Gly–Gly and Ala–Ala as well would give four.)